

TypeScript Hands-On Exercises

1. Data Types & Variable Declaration

Exercise 1 – Declare Variables

Create variables for:

- your name
- age
- isTester
- salary
- city

Exercise 2 – Employee Information

Store employee details and print them using template literals.

2. Primitive Types, Type Annotations & Type Inference

Exercise 1 – Explicit Type Annotation

Create variables using explicit type annotations.

Exercise 2 – Type Inference

Create variables without annotations and let TypeScript infer the type.

Exercise 3 – Fix Type Errors

Fix incorrect type assignments.

3. String, Math, Date Functions

String Exercises:

- uppercase
- lowercase
- substring
- replace
- includes

Math Exercises:

- calculator operations
- random number generation

Date Exercises:

- current date
- month/year
- age calculator

4. If Conditions

Exercise 1 – Voting Eligibility

Exercise 2 – Even or Odd

Exercise 3 – Largest of 3 Numbers

Exercise 4 – Grade Calculator

5. Switch Case

Exercise 1 – Day of Week

Exercise 2 – Calculator using Switch

6. Loops

For Loop:

- print numbers
- multiplication table

While Loop:

- sum of numbers

Do While:

- password retry simulation

Nested Loop:

- star pattern

7. Arrays, Tuples & Enums

Arrays:

- student marks
- array operations

Tuples:

- employee tuple

Enums:

- order status enum

8. Unions, Intersections & Enums

Union Types:

- multi-type variable
- function with union types

Intersection Types:

- employee & manager interfaces

Enums:

- user roles

9. Functions & Interfaces

Functions:

- add function
- arrow function
- optional parameters
- default parameters

Interfaces:

- student interface
- login interface
- interface inheritance

10. Mini Projects

Project 1 – Student Management System

Project 2 – Banking Application

Project 3 – Employee Payroll System

11. Bonus Challenges

- Find duplicate elements in an array
- Palindrome checker
- Count vowels in a string
- Fibonacci series
- Sort array without built-in sort()

These hands-on exercises are designed to help beginners and intermediate learners practice TypeScript concepts effectively through real coding scenarios.